

Strategic Repurposing of Existing Oil Pipeline for CO₂ gas-phase Transport: A Feasibility Framework.

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1 ABSTRACT

Carbon capture, utilization, and storage (CCUS) plays an important role in achieving long-term climate targets, but its large-scale deployment depends on the availability of safe, reliable, and cost-effective CO₂ transport infrastructure. While new pipelines will be required, repurposing existing gas and oil pipelines offers a pragmatic opportunity to accelerate the development of early CO₂ transport networks. In particular, gas-phase CO₂ transport can be an attractive repurposing option where pressure limitations, material properties, or infrastructure constraints restrict dense-phase operation.

This paper presents a structured feasibility study as a critical early decision gate for repurposing existing pipelines to gas-phase CO₂ service. The approach demonstrates how early definition of value-chain premises including CO₂ sources and sinks, transporting volumes, operating envelopes, and CO₂ specification establishes the design basis for hydraulic, safety, and integrity assessments.

In addition to technical considerations, the feasibility phase can also clarify the expected cost implications of repurposing by identifying the required repairs and modifications, as well as the influence of CO₂ specification on processing and operational expenditure. These cost elements provide the foundation for estimating the preliminary Levelized Cost of Transport (LCOT), which serves as an early indicator of the potential CO₂ transport tariff. By filtering technical, safety, and cost-related risks before detailed engineering, the feasibility study reduces uncertainty and supports informed decision-making for pipeline repurposing to CO₂ service.

2 INTRODUCTION

2.1 CO₂ TRANSPORT CHALLENGE AND THE ROLE OF PIPELINE REPURPOSING

The large-scale deployment of carbon capture, utilization, and storage (CCUS) requires the development of reliable transport infrastructure capable of moving significant CO₂ volumes from emission sources to storage sites or utilization hubs. Pipeline transport is widely recognized as the safest and most efficient solution for sustained, high-volume CO₂ transport over long distances. However, the time, cost, and permitting requirements associated with new pipeline construction present a major barrier to rapid CCUS deployment.

Repurposing existing gas and oil pipelines offers a pragmatic opportunity to accelerate the development of CO₂ transport networks, particularly in regions with mature infrastructure. Many existing pipelines are strategically routed between industrial clusters and potential storage locations, making them well suited to form the backbone of early CCUS networks. By reusing existing assets, project developers can reduce capital expenditure, shorten development timelines, and limit environmental and societal impacts compared to new-build projects.

2.2 RELEVANCE OF GAS-PHASE CO₂ TRANSPORT FOR EXISTING PIPELINES

While dense-phase CO₂ transport is often preferred for large-scale, long-distance transmission due to its higher capacity, gas-phase CO₂ transport can represent a viable and attractive alternative for

pipeline repurposing projects. Many legacy pipelines were originally designed for natural gas service with maximum allowable operating pressures that are insufficient for dense-phase CO₂ operation. In such cases, gas-phase transport enables continued use of the existing pipeline without extensive structural upgrades.

Gas-phase CO₂ transport is particularly relevant during early project phases, where captured CO₂ volumes may be limited and increase gradually as additional emitters are connected. Although gas-phase operation results in lower transport capacity, it can support phased network development. However, the change from gas and oil service to CO₂ introduces specific technical, safety, and integrity challenges related to CO₂ properties which must be thoroughly assessed.

2.3 OBJECTIVE AND SCOPE OF THIS PAPER

This paper focuses on the feasibility study as a critical early decision gate for repurposing existing pipelines to gas-phase CO₂ service. In particular, it addresses the Initiation and Screening phases of the repurposing process, where high-level technical, safety, and integrity considerations are evaluated to identify key constraints and potential showstoppers. The paper demonstrates how feasibility-level assessments can provide a robust foundation for subsequent design and qualification activities.

3 FEASIBILITY STUDY AS KEY DECISION GATE FOR CO₂ PIPELINE REPURPOSING

The feasibility phase is the critical starting point of any project aiming to repurpose an existing pipeline, because it establishes the design premises that govern all qualification, engineering, and modification decisions. In repurposing, the pipeline is evaluated under new operating conditions in line with CO₂ pipeline standards for concept and operation. By resolving key uncertainties early, the feasibility phase prevents “designing on the wrong assumptions” and reduces the risk of late-stage rework.

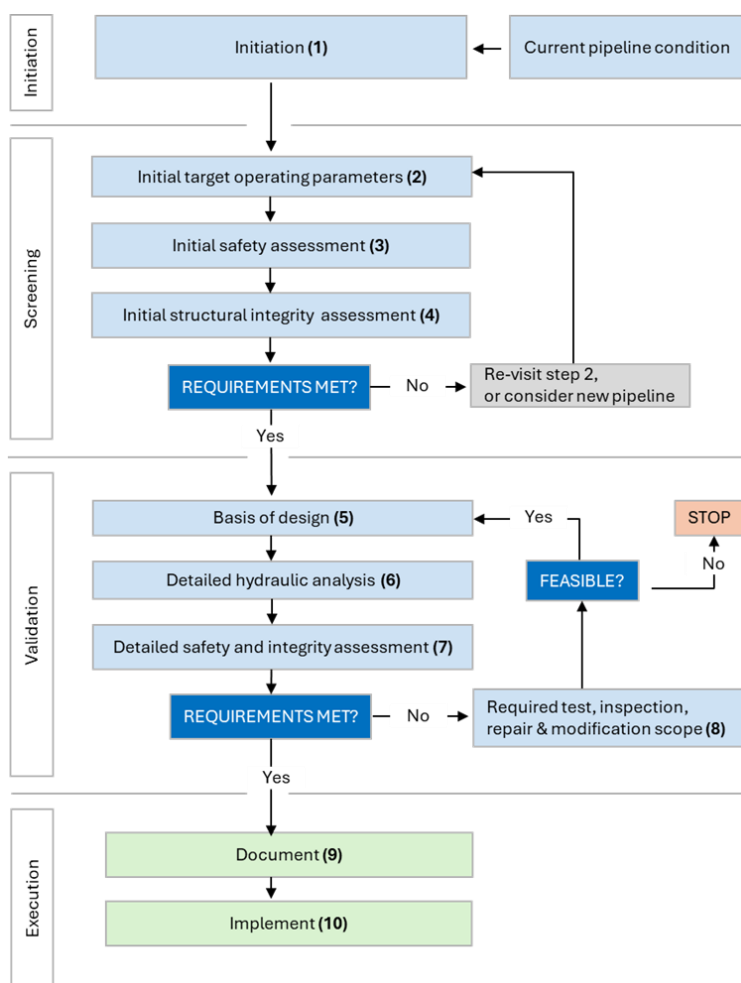
A robust feasibility assessment begins by clarifying the intended CO₂ purpose and establishing operating envelopes, mapping the value-chain boundary conditions (emitters, offtake/storage locations), and establishing expected transport volumes over time. These inputs are then used to evaluate feasible transport conditions based on the existing pipeline’s design and condition. In parallel, the feasibility phase defines the CO₂ specification, including purity requirements and impurity limits. This specification becomes a foundational input for integrity and risk evaluations.

The hydraulic and capacity checks shall be performed to confirm that the repurposed system can realistically meet target throughput and remain within a stable operating envelope along the route. In addition, feasibility screening shall address high-level routing, permitting, and regulatory constraints, and identifies potential integrity and safety showstoppers that could challenge repurposing process. For CO₂ service, particular attention should be on including corrosion threats under adverse impurity/water conditions, fracture control considerations (especially for dense-phase operation), and the potential need for modifications such as isolation/line-break valves, monitoring/detection strategies, and material compatibility upgrades for components exposed to CO₂.

4 DNV STRUCTURED REPURPOSING PROCESS

Repurposing is not a simple “change of medium”; it is a decision process that helps to reduce uncertainties, confirm the technical basis, and define the required modifications and verification activities. In this context, the feasibility study functions as the key early gate: it establishes the design premises (CO₂ purpose, volumes, phase selection, specifications, and early capacity/routing constraints) that determine whether repurposing is feasible. DNV’s has developed a repurposing process that provides a practical framework for converting existing pipelines to CO₂ service, following a structured, phase-based methodology that supports consistent decision-making from concept selection through implementation. The process consist of four phases and is shown below.

Figure 1: DNV repurposing process (DNV)



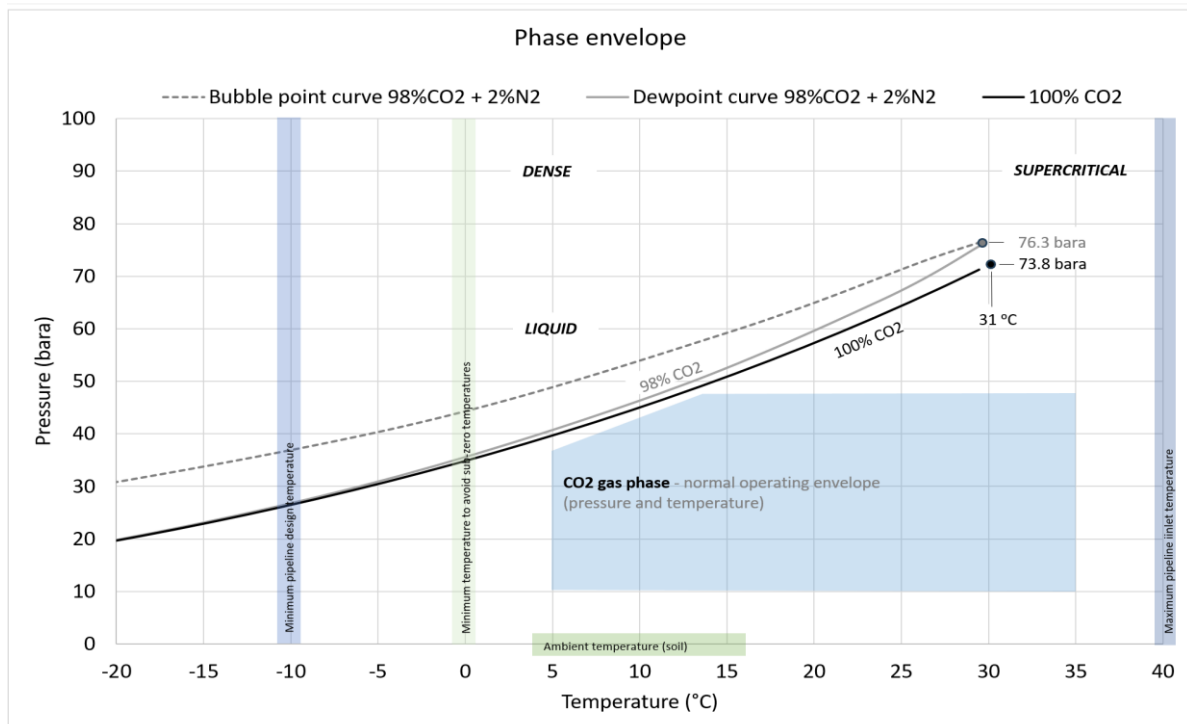
- Initiation phase. The initiation phase covers collection and confirmation of necessary information to start the re-qualification process. In this phase the a candidate pipeline system is identified and the key boundary conditions and premises for repurposing are established (e.g., the intended operating envelope and other fundamental assumptions that will govern subsequent assessments).

- Screening phase. This phase delivers a high-level feasibility evaluation to confirm whether the system is suitable in principle, with emphasis on safety and integrity and with the primary aim of providing high-level guidance on the repurposing pathway and its main decision points.
- Validation phase. If the concept remains viable, the validation phase develops the final technical basis for repurposing through a more detailed assessment of the selected pipeline system, confirming the scope of required testing, inspections, repairs, and modifications needed to meet the defined requirements.
- Execution phase. It covers implementation and close-out, including completion of the modification work and the formal documentation necessary to demonstrate that the repurposed system conforms to the agreed basis and scope.

5 DEFINITION OF OPERATING PREMISES AND CO₂ PHASE ENVELOPE

Before starting with the feasibility step, an initial operating envelope for CO₂, including the expected operating pressure regime must be defined. This early framing is important because it sets the basis for subsequent assessments of hydraulic feasibility, safety consequences, and integrity risks. Carbon dioxide can be transported through pipelines either as a dense liquid or as a gas. Figure 2 illustrates the typical pressure and temperature limits for pipeline transport of gas phase CO₂, both for pure CO₂ and for mixtures containing 2% non-condensable components such as nitrogen (N₂).

Figure 2: For illustration; operating envelope - CO₂ transport in gas phase (DNV)



Based on the maximum design and operating pressures of the existing pipeline system, the feasibility of operating the pipeline under different CO₂ phase conditions can be evaluated, enabling identification of the most suitable operating phase. Existing gas and oil pipelines, typically designed for a Maximum Allowable Operating Pressure (MAOP) in the range of 40–60 bar, will be generally compatible with CO₂ transport in the gaseous phase.

6 IDENTIFICATION AND ASSESSMENT OF CO₂ EMITTERS

The identification and integration of CO₂ emission sources is a critical step following confirmation of the technical feasibility of repurposing an existing pipeline for CO₂ service. Beyond establishing whether sufficient CO₂ volumes can be captured and transported, this step directly influences pipeline utilization, economic viability, and long-term operational robustness. A clear understanding of emitter locations, emission profiles, and expected volume evolution is therefore essential for defining a credible repurposing concept and supporting investment decisions by reducing uncertainty related to throughput, scalability, and revenue stability.

6.1 EMITTER SCREENING AND VOLUME EVOLUTION

The process begins with a high-level screening of potential CO₂ emitters located within a technically and economically viable distance from the existing pipeline corridor. Candidate sources are identified based on emission magnitude, proximity, capture readiness, and expected operational lifetime. For each shortlisted emitter, the volume of CO₂ that could realistically be captured under current or foreseeable operating conditions is estimated, forming the basis for assessing transport scenarios and phased development scenarios.

Defining achievable transport volumes is especially important for repurposed pipelines, as the existing pipeline characteristics may limit initial throughput, while utilization can increase over time as more emitters are connected. These profiles directly affect hydraulic feasibility, compressor requirements, and the selection of the base operating scenario for repurposing. Early alignment between anticipated CO₂ supply and pipeline capacity helps avoid over- or under-design and supports a staged implementation strategy.

6.2 IMPACT OF MULTIPLE SOURCES ON CO₂ COMPOSITION

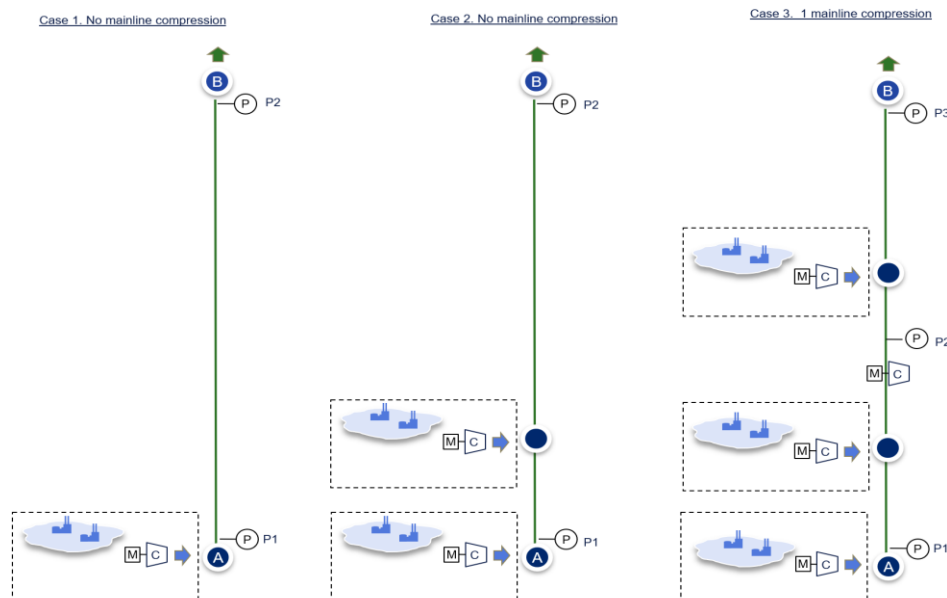
While historically CO₂ pipelines have often transported a relatively uniform product, CCUS networks increasingly rely on the integration of multiple industrial sources. CO₂ streams from cement plants, refineries, biomass facilities, steel manufacturing, or power generation may differ significantly in composition, particularly with respect to trace impurities such as NO_x, SO_x, H₂S, and oxygen. As multiple emitters are connected to a shared pipeline system, the resulting CO₂ composition may fluctuate spatially and temporally along the route. These variations introduce new challenges compared to traditional single-source CO₂ transport and must be explicitly considered during feasibility. The combined effect of multiple sources determines the bounding CO₂ specification that the pipeline must safely accommodate and therefore becomes a key input to integrity assessment and repurposing.

7 HYDRAULIC FEASIBILITY AND CAPACITY ASSESSMENT

In this step, it is essential to verify whether the pipeline system can reliably transport the anticipated CO₂ volumes, assuming an optimal distribution of emitters and a pipeline condition suitable for repurposing. Multiple transport scenarios may be evaluated to determine the feasible hydraulic capacity of the system. Key factors influencing CO₂ transport capacity include:

- Maximum allowable operating pressure, as defined by the current pipeline condition.
- Location of CO₂ entry points, determined by the geographical distribution of emitters.
- Need for intermediate compressor stations

Figure 3: For illustration; Example of Hydraulic assessment (DNV)



Based on the evaluated scenarios, the selected transport configuration provides the basis for subsequent safety and integrity assessment.

8 INITIAL SAFETY AND INTEGRITY ASSESSMENT FOR CO₂ GAS PHASE SERVICE

Once the baseline transport scenario and overall repurposing strategy have been established, the next step is to assess the pipeline's current condition and the potential threats associated with CO₂ service. At the feasibility stage, the objective is to understand how the change in transported product influences pipeline integrity and safety performance, and to identify the key constraints and mitigation measures required to enable safe repurposing and long-term operation. As part of this assessment, indicative dispersion and consequence modelling may be performed to support understanding of the CO₂ release behaviour and the resulting risk profile. Such screening-level modelling supports

understanding of how the overall risk profile changes when transitioning from oil or natural gas service to CO₂ service, and helps identify whether additional or modified mitigation measures are required.

8.1 BASELINE PIPELINE INTEGRITY

A key initial task in the integrity assessment is to establish the baseline condition of the existing pipeline using available integrity data. This includes a systematic review of historical inspection and maintenance records, original design specifications, material properties, and known defect types. Existing defects within the pipeline should be identified and assessed. The assessment should consider defects such as internal and external corrosion, metal loss, cracking, and weld-related imperfections, and should evaluate their acceptability under the proposed CO₂ operating conditions over the intended service life.

In parallel, location-specific threats shall be reviewed, including third-party interference, geotechnical hazards, and the presence and performance of previous repairs. Establishing a clear understanding of the pipeline's current integrity status is essential, as it defines the boundary conditions within which CO₂ transport can be considered and provides the reference point for evaluating the impact of the change in service.

8.2 CO₂-SPECIFIC THREATS

At this stage, it is essential to understand the implications of changing service conditions. This includes evaluation of potential new threats and risks and considerations necessary to ensure safe repurposing and long-term operational reliability. The outcome of such assessments is typically a structured set of recommendations addressing CO₂ specifications, operational procedures, emergency response arrangements, integrity management requirements, and overall compliance considerations. These recommendations form key inputs to the development of technical specifications, detailed design, and operational planning for CO₂ pipeline service.

8.3 COMPARATIVE SAFETY RISK: CO₂ VERSUS OIL/NATURAL GAS

Following the identification of applicable threats and the evaluation of baseline integrity and CO₂ fluid properties, a comparative safety risk assessment should be performed to understand how the transition from gas and oil service to CO₂ affects overall safety performance. This comparison should focus on changes in both failure likelihood and consequence profiles.

Key aspects include potential shifts in dominant failure mechanisms such as an increased relevance of internal corrosion or fracture-related threats as well as changes in release consequences. Understanding these differences is essential for determining whether the target safety level can be achieved or whether additional safeguards, operational controls, or emergency response measures are required for CO₂ service.

8.4 FRACTURE AND DECOMPRESSION CONSIDERATIONS

For pipelines transporting CO₂ in the gas phase, running ductile fracture requires less strict material properties for arrest. Nevertheless, a screening-level fracture propagation assessment remains

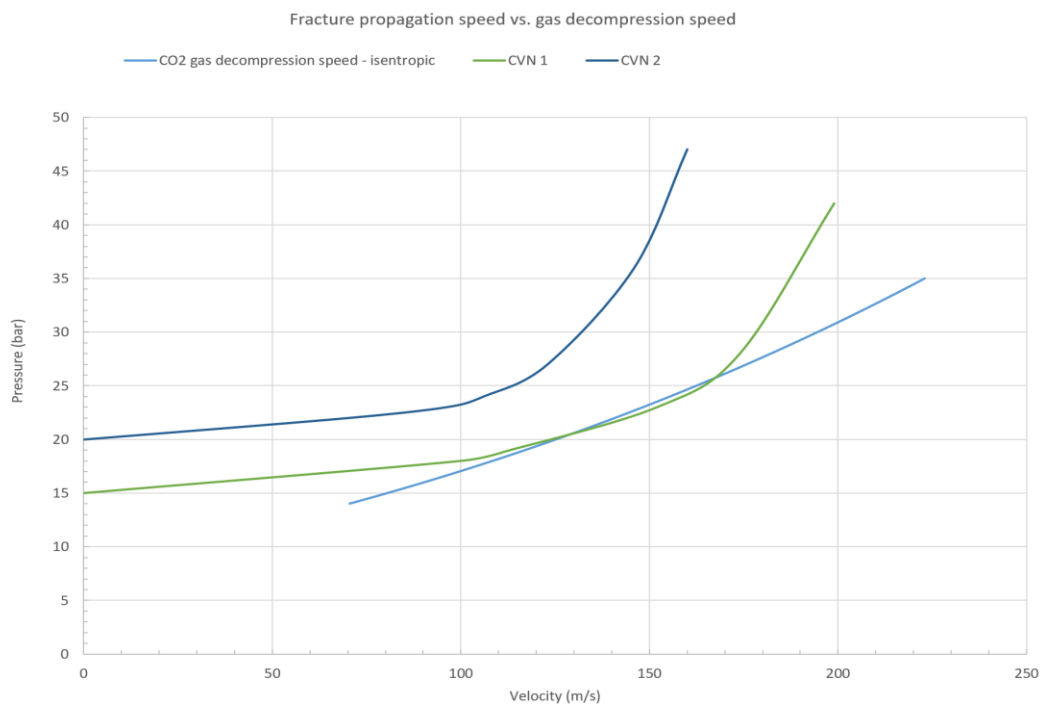
advisable to verify that the material toughness is adequate to arrest a potential running fracture. Modern new-build pipelines are typically produced from steels with high and well-controlled toughness properties, which substantially reduces the likelihood of fracture propagation. Older pipeline steels may exhibit lower Charpy V-Notch (CVN) impact energy values and reduced fracture toughness relative to current new-build pipelines.

An assessment can be performed using the Battelle Two-Curve Method to confirm fracture arrest capability. The Battelle Institute developed a generalized method for predicting whether a fracture in natural gas pipelines will arrest or propagate. Although the thermodynamic properties of gaseous CO₂ differ from those of natural gas, both fluids behave as single-phase compressible gases under the operating conditions considered. The fundamental physics governing the interaction between fluid decompression and crack propagation therefore remains unchanged. Consequently, the Battelle two-curve method may be considered applicable. In this method, two curves are developed:

- The decompression curve, representing the gas pressure decay behind a propagating crack.
- The fracture propagation curve, representing the material's resistance to fracture, is based on Charpy V-notch (CVN) energy.

The diagram below presents an illustrative fracture control screening assessment for a representative pipeline diameter and wall-thickness combination, comparing CO₂ gas decompression behaviour with fracture propagation speed for two different Charpy V-Notch (CVN) toughness values. A pipeline is considered resistant to running fracture if the material toughness curve lies above the decompression curve.

Figure 4: For illustration; Example of fracture control screening assessment (DNV)



As shown in the figure above, this condition is satisfied for both CVN 2 value under the illustrated operating conditions, indicating adequate fracture arrest capability. However, for the lower toughness

case (green line, CVN 1), the fracture propagation curve crosses the decompression curve, indicating its fracture control limitation.

9 MODIFICATION AND REPURPOSING SCOPE

Based on the outcomes of the feasibility study and the initial safety and integrity assessment, the scope of required modifications and repurposing activities can be defined. The identified integrity threats and safety implications associated with CO₂ service directly inform the type and extent of measures. These measures may include repairs to address existing integrity deficiencies, replacement or upgrading of equipment to ensure material compatibility with CO₂, and the installation of additional isolation and safety systems. Furthermore, the repurposing to CO₂ service requires enhanced monitoring, control, and leak detection capabilities. Emergency response procedures and operational philosophies must also be reviewed and updated to reflect the hazard profile associated with CO₂ releases.

The resulting modification and repurposing scope represents not only a translation of identified risks into concrete technical and operational measures, but also a key input to the techno-economic evaluation of the repurposing concept. The results of the feasibility study establishes the technical basis for cost estimation and comparison with alternative transport solutions, thereby supporting development of a robust business case for pipeline repurposing to CO₂ service.

10 COST DRIVERS AND ECONOMIC IMPLICATIONS

An evaluation of repurposing an existing pipeline for CO₂ transport can also include an early assessment of the expected cost drivers and economic implications. While repurposing generally offers a significant cost advantage compared to constructing new CO₂ pipelines, the total investment required can vary widely depending on the existing pipeline condition and required modifications. The feasibility study therefore serves not only as a technical decision gate but also as a critical input to developing a credible cost baseline that supports the overall business case.

Repurposing avoids major cost components associated with new-build pipelines, such as right-of-way acquisition, new permitting, extensive civil works, and routing constraints, allowing a reduction of CAPEX and shortening of delivery timelines. However, these advantages depend heavily on the technical feasibility established early in the repurposing process. Pipelines that require substantial repairs, widespread component upgrades, or major system modifications may approach cost levels similar to new construction, eroding the repurposing benefit.

Repurposing costs are primarily influenced by how much testing, verification, and inspection is required to re-qualify the pipeline and confirm its baseline condition, the extent of integrity repairs necessary to address existing defects or CO₂-specific risks, and the system modifications required to accommodate CO₂ operations. For example, valve replacements, additional isolation valves, monitoring upgrades, and compression systems where needed all factor into the overall cost impact. Long-term operational expenses also factor into overall cost considerations, including energy demand for aspects like compression, routine integrity inspections, equipment replacement and leak detection.

Re-commissioning the pipeline, such as conducting pipeline cleaning, performing a new hydrotest followed by dewatering and drying, can also represent a significant portion of the overall repurposing cost. Together, these elements highlight how the feasibility phase defines the expected cost envelope by establishing the scope and scale of technical modifications.

Beyond capital and operating costs, repurposing studies often incorporate the LCOT as a preliminary economic indicator that expresses the transport cost per tonne of CO₂ over the full asset lifetime (typically stated in EUR/t). LCOT provides an early basis for estimating the potential CO₂ transport tariff by linking technical requirements, such as expected volume profiles, operating pressure regimes, and the extent of required modifications, to long-term economic performance. Establishing these parameters during the feasibility phase enables developers to form an early tariff outlook, supporting commercial planning and alignment across emitters, transport operators, and storage providers.

By reducing uncertainty around operating envelopes, impurity limits, pipeline condition, and the scope of required modifications, the feasibility study enables the development of a realistic techno-economic assessment for the repurposing concept. The resulting early cost baseline supports the preliminary estimation of a potential CO₂ transport tariff, reflecting how technical requirements and expected throughput influence long-term economic performance. Even when technically feasible, a repurposing concept may face economic constraints, and early visibility of these cost drivers ensures that subsequent technical and commercial decisions are both technically robust and economically justified.

11 CONCLUSION

A structured feasibility study provides an effective early decision gate for repurposing existing pipelines to CO₂ service by filtering technical, safety, and integrity risks before detailed engineering is initiated.

Gas-phase CO₂ transport represents a pragmatic and near-term repurposing option for many existing gas and oil pipelines, particularly where pressure limitations and infrastructure constraints prevent dense-phase operation.

Early definition of operating premises and CO₂ phase selection is critical, as these parameters govern hydraulic feasibility, integrity requirements and safety consequences.

The CO₂ specification, including water content and impurity limits, is a key integrity driver and directly influences internal corrosion risk and operational constraints.

Repurposing to CO₂ service introduces potential integrity and safety challenges compared to gas and oil service, such as increased relevance of internal corrosion and consequence profiles.

Screening-level safety and integrity assessments enable systematic identification of CO₂-specific risks and support development of targeted mitigation measures.

The resulting modification and re-qualification scope represents a direct translation of identified risks into concrete technical and operational measures.

This risk-driven scope forms the technical basis for cost estimation and techno-economic evaluation, supporting comparison with alternative transport solutions and enabling informed business decisions on pipeline repurposing to CO₂ service.

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